
Amazon Redshift – Complete Overview

What is Amazon Redshift?

Amazon Redshift is a **fully managed, petabyte-scale data warehouse** service in the cloud.

It is designed for **analytical workloads** – like **reporting, dashboards, business intelligence, and big data analytics**.

 Think of it like:

A high-performance SQL-based database designed to analyze **millions to billions** of rows of data **very fast**.

Key Concepts

Concept	Description
 OLAP	Redshift is built for Online Analytical Processing (OLAP), not for OLTP (like RDS).
 Columnar Storage	Stores data by column rather than row. Greatly improves compression and speed for analytics.
 Massively Parallel Processing (MPP)	Distributes queries and data across multiple nodes to process in parallel.
 SQL Interface	You can use familiar PostgreSQL-like SQL to query Redshift.
 Data Loading	Easily load data from S3, DynamoDB, RDS, or via AWS Glue & Kinesis .

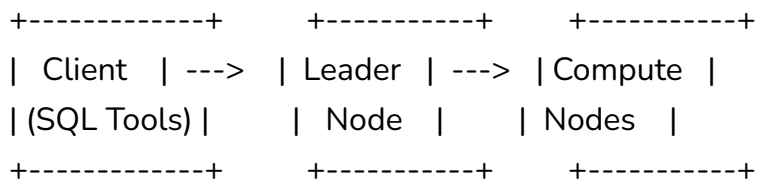


Security

Supports **VPC, IAM, KMS encryption, audit logging,** and more.



Architecture



- **Leader Node:** Accepts queries, creates execution plan, coordinates execution.
 - **Compute Nodes:** Execute queries in parallel; store actual data.
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Core Components

Component	Description
 Leader Node	Manages query parsing and optimization
 Compute Nodes	Perform actual data processing
 Node Types	Dense Storage (DS2) or Dense Compute (DC2/RA3)
 Clusters	Collection of Redshift nodes
 Slices	Each compute node is divided into slices (unit of parallelism)
 Distribution Styles	How data is spread across nodes (KEY, EVEN, ALL)



Sort Keys

Help optimize queries by storing data in a defined sort order



COPY Command

Used to load data into Redshift from S3, DynamoDB, etc.



Security Features

-  **Encryption** at rest (KMS) and in transit (SSL)
 -  **IAM Roles & Policies**
 -  **Audit Logging** with CloudTrail
 -  **VPC Support**
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Integration with AWS

Service	Integration
 Amazon S3	Load/export data using COPY/UNLOAD
 Amazon Athena	Query same datasets via Athena & Redshift
 AWS Glue	Data catalog & ETL integration
 Kinesis	Stream real-time data into Redshift
 Redshift Spectrum	Query data stored directly in S3 without loading into Redshift



Common Use Cases

-  Business Intelligence (BI) Dashboards
 -  Ad-hoc Analytics
 -  Complex Aggregations on Large Datasets
 -  Data Lake Queries (with Redshift Spectrum)
 -  Machine Learning Data Prep (for SageMaker)
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Performance Features

-  **Columnar compression** reduces disk I/O
 -  **Concurrency scaling** (adds transient clusters automatically)
 -  **Materialized views** for faster query performance
 -  **Result caching**
 -  **Workload Management (WLM)**: Assigns priority queues to queries
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Pricing Model

You are billed for:

- **Node type** and **node count** (on-demand or reserved)
- **RA3 with managed storage** lets you scale compute and storage **independently**

- Additional charges for **Spectrum queries** and **data transfers**

 Example: COPY Data from S3

```
COPY users
FROM 's3://my-bucket/users.csv'
CREDENTIALS 'aws_iam_role=arn:aws:iam::123456789012:role/MyRedshiftRole'
CSV;
```

 Summary Table

Feature	Detail
Storage	Columnar
Engine	Massively Parallel
Query Language	PostgreSQL (SQL)
Data Sources	S3, RDS, DynamoDB, etc.
Use Case	Analytics, BI, Big Data
Alt to	Snowflake, BigQuery, Azure Synapse
